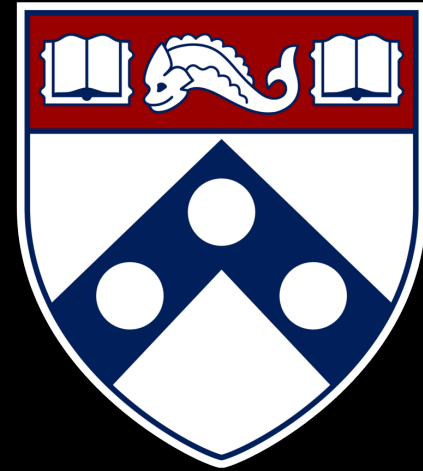


Empirical Security & Privacy, for Humans



UPenn CIS 7000-010
9/11/2025



Security – Passwords

Readings

(optional)

Do Users' Perceptions of Password Security Match Reality?

Blase Ur, Jonathan Bees[†], Sean M. Segreti, Lujo Bauer, Nicolas Christin, Lorrie Faith Cranor
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ABSTRACT

Although many users create predictable passwords, the extent to which users realize these passwords are predictable is not well understood. We investigate the relationship between users' perceptions of the strength of specific passwords and their actual strength. In this 165-participant online study, we ask participants to rate the comparative security of carefully juxtaposed pairs of passwords, as well as the security and memorability of both existing passwords and common password-creation strategies. Participants had serious misconceptions about the impact of basing passwords on common phrases and including digits and keyboard patterns in passwords. However, in most other cases, participants' perceptions of what characteristics make a password secure were consistent with the performance of current password-cracking tools. We find large variance in participants' understanding of how passwords may be attacked, potentially explaining why users nonetheless make predictable passwords. We conclude with design directions for helping users make better passwords.

Author Keywords

User behavior; perceptions of security; passwords; authentication; users' folk models; usable security

ACM Classification Keywords

H.5.m Information Interfaces and Presentation (e.g., HCI): Miscellaneous; K.6.5 Security and Protection: Authentication

INTRODUCTION

For better or worse, passwords remain today's dominant form of user authentication [11]. While the predictability of user-chosen passwords has been widely documented [9, 37, 51, 72, 74, 77, 80], very little research has investigated users' perceptions of password security. That is, do users realize they are selecting terrible passwords and choose to do so intentionally, or are they unwittingly creating weak passwords when they

chosen to exhibit particular characteristics, as well as common strategies for password creation and management. We compare participants' perceptions to the passwords' actual resilience to a variety of large-scale password-guessing attacks.

In the first of four tasks, we showed participants 25 pairs of passwords differing in specific characteristics (e.g., appending a digit, as opposed to a letter, to the end of the password). We asked participants to rate which password was more secure, if any, and to justify their rating in free text. In the second and third tasks, we showed participants a selection of passwords from the well-studied breach of the website RockYou [72], as well as descriptions of common password-creation strategies. We asked participants to rate both the security and the memorability of each password or strategy. In the fourth task, we had participants articulate their model of password attackers and their expectations for how attackers try to guess passwords.

We observed some serious misconceptions about password security. Many participants overestimated the benefits of including digits, as opposed to other characters, in a password. Many participants also underestimated the poor security properties of building a password around common keyboard patterns and common phrases. In most other cases, however, participants' perceptions of what characteristics make a password more secure matched the performance of today's password-cracking tools. This result calls into question why users often fail to follow their (correct) understanding when crafting passwords. However, most participants displayed an unrealistic mental model of attackers, which may prevent them from fully accounting for the actual spectrum of threats to their passwords.

Although much has been written about text passwords in recent years, our study is the first to focus specifically on users' perceptions of security. The main outcome of our work is to inform design directions for helping users both make stronger passwords and better understand the implications of their password-creation decisions.

A Two-Decade Retrospective Analysis of a University's Vulnerability to Attacks Exploiting Reused Passwords

Alexandra Nisenoff^{†*}, Maximilian Golla^{‡‡}, Miranda Wei^{†*}, Juliette Hainline[†], Hayley Szymanek[†], Annika Braun[†], Annika Hildebrandt[†], Blair Christensen[†], David Langenberg[†], Blase Ur[†]
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[‡] Max Planck Institute for Security and Privacy, [‡] University of Washington

Abstract

Credential-guessing attacks often exploit passwords that were reused across a user's online accounts. To learn how organizations can better protect users, we retrospectively analyzed our university's vulnerability to credential-guessing attacks across twenty years. Given a list of university usernames, we searched for matches in both data breaches from hundreds of websites and a dozen large compilations of breaches. After cracking hashed passwords and tweaking guesses, we successfully guessed passwords for 32.0% of accounts matched to a university email address in a data breach, as well as 6.5% of accounts where the username (but not necessarily the domain) matched. Many of these accounts remained vulnerable for years after the breached data was leaked, and passwords found verbatim in breaches were nearly four times as likely to have been exploited (i.e., suspicious account activity was observed) than tweaked guesses. Over 70 different data breaches and various username-matching strategies bootstrapped correct guesses. In surveys of 40 users whose passwords we guessed, many users were unaware of the risks to their university account or that their credentials had been breached. This analysis of password reuse at our university provides pragmatic advice for organizations to protect accounts.

1 Introduction

Despite their disadvantages, passwords remain widely used for authentication [7]. Organizations must protect against large-scale attacks on users' passwords. An adversary may leverage **reused passwords**—when the same individual picks similar or identical passwords for different services [10, 80] to cope with having to remember numerous passwords [16]. If any one of these services suffers a data breach, attackers

The ability to conduct attacks that exploit reused password has increased as hundreds of websites have had their password databases stolen and leaked over the last decade [34]. We term the breach of a single service an **individual service breach**. In recent years, hackers have also packaged credentials from many different services into **breach compilations** containing hundreds of millions or even billions of credentials [24].

To protect an organization against attacks exploiting common passwords, system administrators can institute straightforward blocklists [25, 70]. Protecting an organization from reused passwords, however, is far more complex. A vulnerable password is specific to one user based on their credentials on other sites at any past or future time. Furthermore, prospective attackers often have far more information than system administrators. Attackers may know about a successful breach that system administrators may not hear about for years, or ever. Further, attackers may pool resources to crack hashes and reveal the plaintext needed for an attack, while the system administrator may be left only with uncracked hashes [9].

In recent years, researchers and practitioners have developed compromised-credential-checking tools to try to defend users. For instance, Chrome [73], Firefox [55], and Safari [12] notify users if their passwords appear in a data breach. The Have I Been Pwned (HIBP) service [32], itself integrated with 1Password [13], enables users to check for their appearance in a data breach. Supporting these efforts, academic work has proposed protocols that underpin compromised-credential-checking tools [40, 43, 44, 59, 82, 83] and sought to improve the usability of data breach notifications [22, 31, 53, 79, 90, 92].

Despite prior work, many questions remain for system administrators trying to protect their organizations from attacks exploiting reused passwords. For what amount of time are accounts vulnerable? Out of hundreds of data breaches, how

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Although much has been written about text passwords in recent years, our study is the first to focus specifically on users' perceptions of security. The main outcome of our work is to inform design directions for helping users both make stronger passwords and better understand the implications of their password-creation decisions.

Password security: Perception vs. reality

- Password predictability in 2016 (and today?) is high. Why?
- Two hypotheses
 - Users unwittingly select predictable passwords because they misunderstand what makes a password guessable
 - Users prioritize memorability, or some other feature, over predictability
- Let's test these hypotheses experimentally!
 - Recruit participants on Amazon Mechanical Turk
 - Give them a small set of technical exercises and free-response questions; takes about 30 minutes
 - Analyze the data, and draw conclusions

Study design: 5 parts

- The first part of the study asked about demographics, including age and gender, and if a security professional
- Second part: Password pair assessment (comparison)
- Third part: Password direct assessment, of security & memorability
- Fourth part: Password creation and management strategy assessment
- Fifth part: Impressions about attackers
 - Done last to avoid priming participants

Part 2: Comparison questions

In your opinion, which of the following passwords is more secure?

punk4life			punkforlife			
punk4life is much more secure	punk4life is more secure	punk4life is slightly more secure	Both passwords are equally secure	punkforlife is slightly more secure	punkforlife is more secure	punkforlife is much more secure
☐	☐	☐	☐	☐	☐	☐

Why? *

25 kinds of difference, users saw one of three password pairs per category
Password pairs selected randomly/systematically from RockYou dataset (real-world PW DB that was leaked)
26th category is an **attention check**: Show the same password
Presented in randomized order, and randomized left/right position

Part 3: Password rating questions

Please rate the **security** of the following password: **rolltide1** *

1 (very insecure)	2	3	4	5	6	7 (very secure)
☐	☐	☐	☐	☐	☐	☐

Please rate the **memorability** of the following password: **rolltide1** *

1 (very hard to remember)	2	3	4	5	6	7 (very easy to remember)
☐	☐	☐	☐	☐	☐	☐

20 passwords selected from the RockYou dataset

Part 4: Strategy rating questions

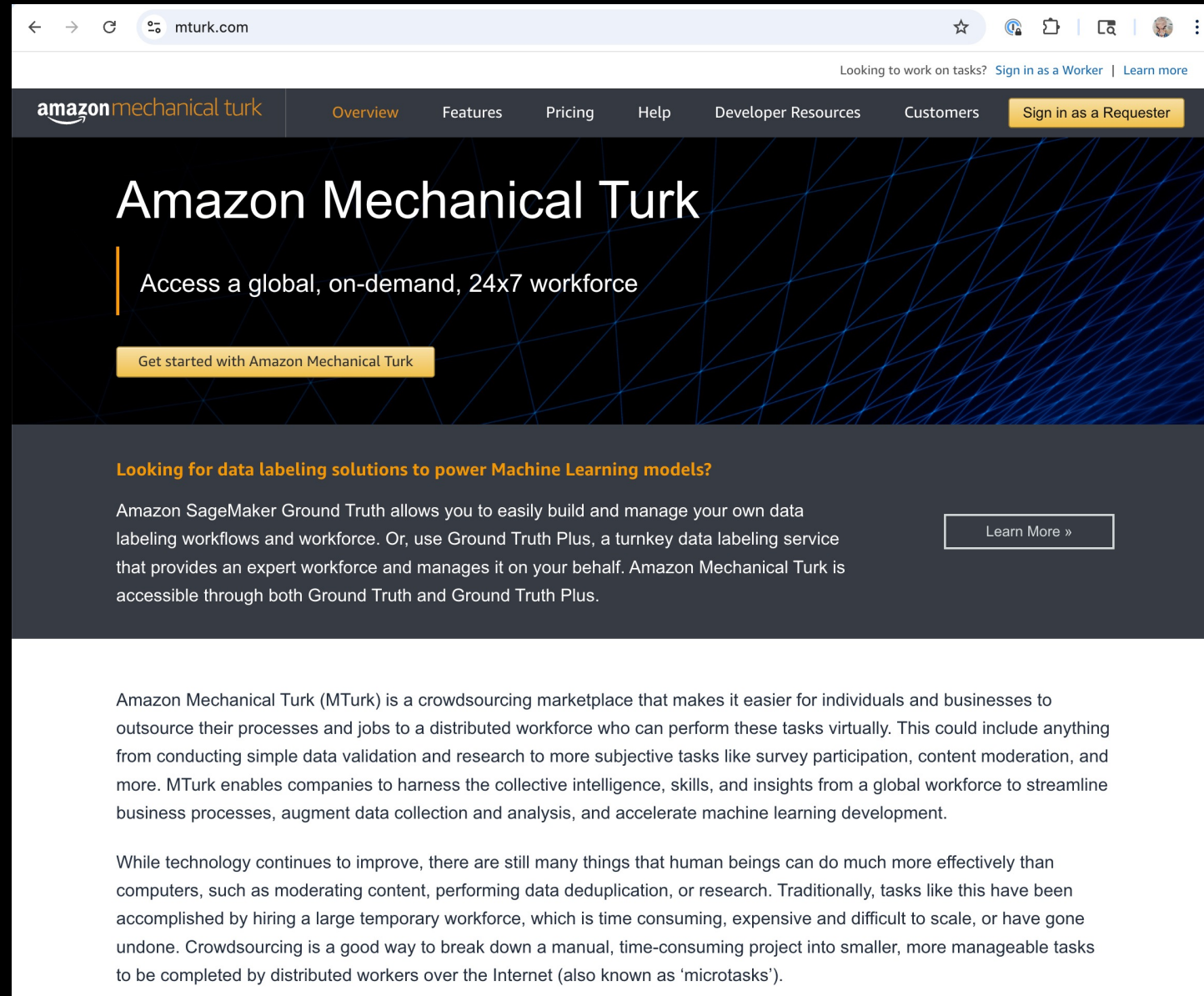
- Similar look to Part 3, but presented the strategy
- Chose common strategies from prior work on password creation and password management
 - Example: create a password “using a phrase taken from the lyrics to a song (e.g., somewhere over the rainbow).”

Part 5: Attacker questions

- 7 questions with open-ended answers
 - “what characteristics make a password {easy, hard} for an attacker to guess?”
 - “describe the type of attacker (or multiple types of attackers), if any, whom you worry might try to guess your password”
 - ...

Study participants

- 165 recruited from Mechanical Turk in USA
- Nearly gender-balanced
- 33 U.S. states.
- Age from 18–66
 - mean age of 34.2 years
 - median of 33
- Few participants had special familiarity with security
- Paid \$5 to participate



The screenshot shows the Amazon Mechanical Turk homepage. At the top, there's a navigation bar with the Amazon Mechanical Turk logo and links for Overview, Features, Pricing, Help, Developer Resources, and Customers. A 'Sign in as a Requester' button is on the right. Below the navigation bar, the main heading is 'Amazon Mechanical Turk' with the tagline 'Access a global, on-demand, 24x7 workforce'. A yellow button says 'Get started with Amazon Mechanical Turk'. Further down, there's a section titled 'Looking for data labeling solutions to power Machine Learning models?' with a paragraph about Amazon SageMaker Ground Truth and a 'Learn More »' button. At the bottom, there's a paragraph explaining what Amazon Mechanical Turk (MTurk) is and how it works.

amazonmechanicalturk Overview Features Pricing Help Developer Resources Customers Sign in as a Requester

Amazon Mechanical Turk

Access a global, on-demand, 24x7 workforce

Get started with Amazon Mechanical Turk

Looking for data labeling solutions to power Machine Learning models?

Amazon SageMaker Ground Truth allows you to easily build and manage your own data labeling workflows and workforce. Or, use Ground Truth Plus, a turnkey data labeling service that provides an expert workforce and manages it on your behalf. Amazon Mechanical Turk is accessible through both Ground Truth and Ground Truth Plus.

Learn More »

Amazon Mechanical Turk (MTurk) is a crowdsourcing marketplace that makes it easier for individuals and businesses to outsource their processes and jobs to a distributed workforce who can perform these tasks virtually. This could include anything from conducting simple data validation and research to more subjective tasks like survey participation, content moderation, and more. MTurk enables companies to harness the collective intelligence, skills, and insights from a global workforce to streamline business processes, augment data collection and analysis, and accelerate machine learning development.

While technology continues to improve, there are still many things that human beings can do much more effectively than computers, such as moderating content, performing data deduplication, or research. Traditionally, tasks like this have been accomplished by hiring a large temporary workforce, which is time consuming, expensive and difficult to scale, or have gone undone. Crowdsourcing is a good way to break down a manual, time-consuming project into smaller, more manageable tasks to be completed by distributed workers over the Internet (also known as 'microtasks').

Qualitative analysis (text responses)

Quantitative analysis (numeric, Boolean data)

The Coding Manual for Qualitative Researchers

»»» 3E «««

Johnny Saldaña

 **SAGE**
Los Angeles | London | New Delhi
Singapore | Washington DC

Experimental Design and Analysis

Howard J. Seltman

July 11, 2018

These free books linked from course webpage

Results: Comparison

59 pairs (79%) were consistent with the performance of current password-cracking approaches. Characterized practices:

- capitalizing the middle of words, rather than the beginning
- putting digits and symbols in the middle of the password, as opposed to the end
- using random-seeming digit sequences, rather than years or obvious sequences
- using symbols in place of digits
- preferring dictionary words over common first names
- avoiding personal content (e.g., a relative's name)
- avoiding terms related to the account (e.g., "survey" for an MTurk password)

Table 4. The 25 hypotheses we investigated among password pairs. The # column indicates for how many of the three pairs per hypothesis participants' perceptions matched the hypothesized perception.

Hypothesized user perception	#
<i>Capitalization</i>	
Non-standard capitalization more secure than capitalizing first letters	3
<i>The use of letters vs. digits vs. symbols</i>	
Appending a lowercase letter more secure than appending a digit	2
Appending lowercase letters more secure than appending digits	0
Symbol more secure than corresponding digit (e.g., "!" vs. "1")	3
All-symbol password more secure than all-digit password	3
Adding an exclamation point makes a password more secure	3
Appending 1 makes a password more secure	3
Appending 1! makes a password more secure	3
<i>Location of digits and symbols</i>	
Digit in middle of password more secure than at beginning or end	3
Symbol in middle of password more secure than at beginning or end	2
<i>Choice of digits and symbols</i>	
Appending random digits more secure than appending a recent year	3
Random digits more secure than common sequence (e.g., "123")	3
<i>Choice of words and phrases</i>	
Dictionary word more secure than a person's name	3
Word that is hard to spell more secure than easy-to-spell word/phrase	1
Uncommon phrase more secure than common phrase	0
Phrase more secure than single word	2
<i>Targeted and personal information</i>	
Unrelated word more secure than word related to the account	3
Unrelated name more secure than name of friend/family	3
Unrelated date more secure than birthdate of friend/family	3
<i>Keyboard patterns</i>	
Common keyboard pattern more secure than common phrase	1
Password without obvious pattern more secure than keyboard pattern	2
<i>Character substitutions</i>	
Lowercase letter less secure than number/symbol (e.g., "3" for "e")	3
Uppercase letter less secure than number/symbol (e.g., "3" for "E")	1
Relevant digit (e.g., "punk4life") less secure than unrelated digit	3
Relevant digit (e.g., "4") less secure than full word (e.g., "for")	0

Results: Comparison

16 pairs (21%) were not consistent.

Misconceptions:

- adding digits inherently makes a password more secure than using only letters
- substituting digits or symbols for letters makes a password more secure
- overestimated the security of keyboard patterns
- misjudged the popularity of particular words and phrases

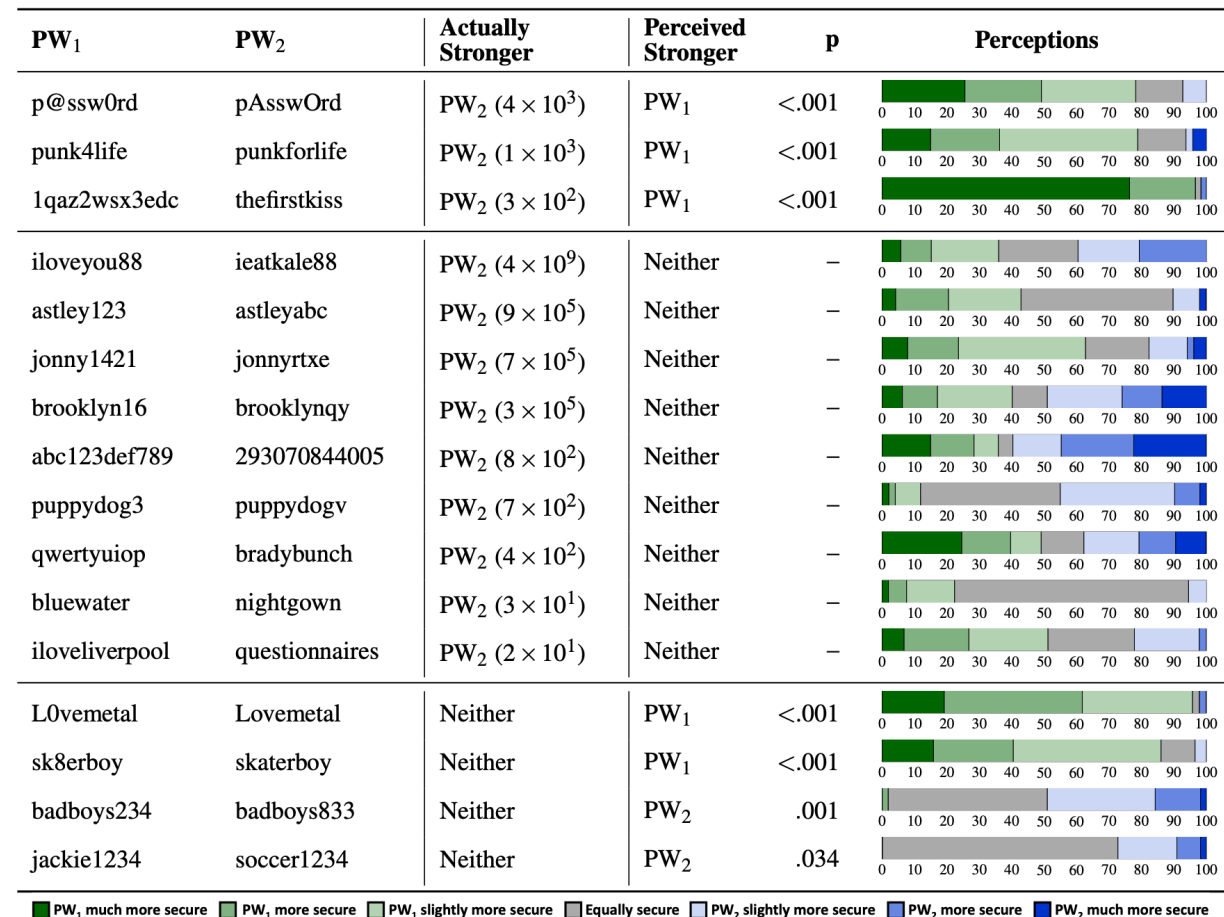


Table 5. Pairs of passwords for which participants' perceptions of the relative security of the passwords differed from actual security. The number in parentheses indicates how many times stronger PW₂ was than PW₁ (ratio of guess numbers).

Results: Direct assessment

Participants' memorability ratings were inversely correlated with strength ratings (Spearman's $\rho = -0.678$, $p < .001$).

Characteristic	Coefficient	Pr(> z)
Length	0.144	<.001
Contains uppercase letter	0.584	.020
Contains digit	0.971	<.001
Contains symbol	1.220	.006
Interaction: Upper*Digit	0.441	.010
Interaction: Digit*Symbol	-0.613	.008

Table 7. Significant terms in our mixed-model, ordinal regression of how password characteristics correlate with participants' *security* ratings.

Characteristic	Coefficient	Pr(> z)
Length	-0.125	<.001
Contains digit	-0.753	<.001

Table 8. The only two significant terms in our mixed-model, ordinal regression of how characteristics correlate with *memorability* ratings.

Results: Strategy

Strategy	ρ	Perceived Security	Perceived Memorability
S1: Starting with a word that comes to mind, and then adding digits or symbols to the end (e.g., bubblegum1!).	-0.22		
S2: Using a phrase taken from the lyrics to a song (e.g., somewhere over the rainbow).	-0.26		
S3: Using a phrase that you make up exclusively for this account and that has nothing to do with the account (e.g., skyscraper cornstalks).	-0.24		
S4: Using the name of one of your family members and their birth year (e.g., Zachary1976), assuming that information is not on Facebook or other social media.	-0.30		
S5: Combining words from two different languages (e.g., desaparecido rainbow).	-0.23		
S6: For your password, using a date that is meaningful to you (e.g., 12151976 because your sibling was born on 12/15/1976), assuming that information is not on Facebook or other social media.	-0.27		
S7: Basing a password on a phrase that describes your relationship to the account (e.g., iloveshoppingonamazon for your Amazon.com password).	-0.47		
S8: Building the password by following a pattern on the keyboard (e.g., 1qaz2wsx3edc).	-0.47		
S9: Using the same password that you use for other accounts.	-0.40		
S10: Using a tool that can randomly generate a complex password for you.	-0.45		
S11: Picking a complex password and writing that password down on a piece of paper that only you know about.	n.s.		

Table 6. Perceptions of the security and memorability of strategies. Participants rated both on a 1–7 scale, where 7 (darker colors on the graphs) indicates “very secure” and “very easy to remember,” respectively. Spearman’s ρ indicates the correlation between security and memorability ratings.

Results: Strategy

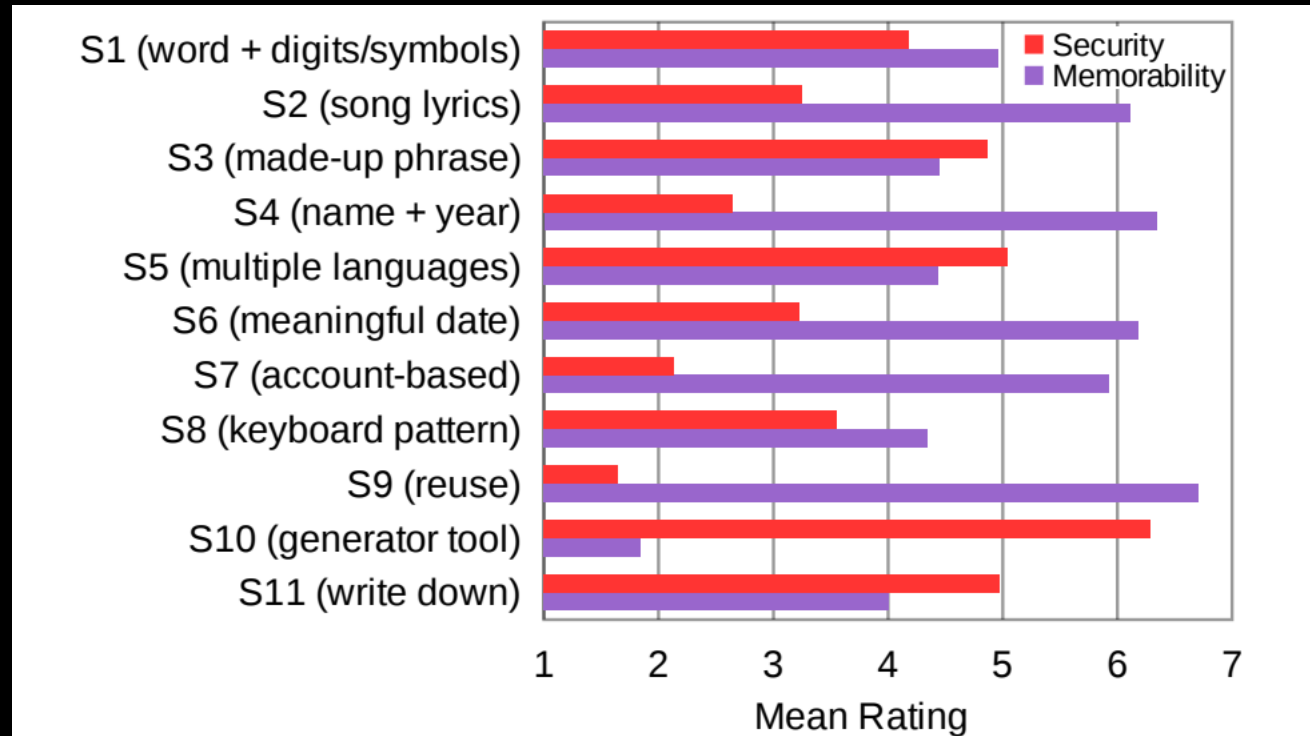


Figure 3. Mean ratings for the security and memorability of the 11 password-creation strategies.

Results: Attacker characterization

Motivation	#	%
Financial payoff	109	66%
Gather personal information	67	41%
Identity theft	33	20%
Fun / prove they can	10	6%
Spamming	6	4%
Spying	4	2%

Table 2. Themes' frequency of occurrence in participants' responses to "why would an attacker try to guess your password, if at all?"

Guessing Method	#	%
Automated, large-scale	121	73%
Software / algorithms	79	48%
"Brute force"	42	26%
Dictionaries / words	27	16%
Common passwords	26	16%
Common names	8	5%
Try guessing dates	7	4%
Targeted to user	72	44%
Use personalized information	62	38%
Social engineering	7	4%
Manual guessing	4	2%
Other means	22	13%
Hacking into system / database	12	7%
Keyloggers	10	6%
Phishing	5	3%

Table 3. Themes' frequency of occurrence in participants' responses to "how do attackers try to guess your password?"

Themes, discussion points

- What are the limitations of this study? Why might its conclusions be wrong?
 - E.g., technical issues with the data analysis, choice of participants, etc.
- User perceptions were mostly in line with good practices. Why, then, are passwords in the wild generally predictable?
- What interventions might improve the situation? How would we test them?

Readings

(optional)

Talk video and PDF of slides at
<https://www.usenix.org/conference/userixsecurity23/presentation/nisenoff-retrospective>

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Summary results

- Guessed 32% of passwords in historical DB by leveraging reuse
 - As compared to 6.5% without considering reuse
 - 35.5% of valid guesses were for current passwords
- Of those guessed by reuse
 - 54.7% were verbatim reuse, vs. 45.3% based on tweaks
- Vulnerability is real
 - Some historical observed exploits seemed to coincide with data breaches
 - Passwords were vulnerable for long after a breach (median of 5 years)

Discussion points

- Ways that the study leveraged the historical password database
- Ethics concerns
- Technical challenges
- What the study might have missed or overstated (limitations)
- Other inferences to draw from results
- Recommendations, and relevance today

Thoughts?

END